

Lab 4: Practice with ggplot2 and tidyverse

! Important

This lab is due on Monday, September 14 at 11:59pm. To be considered on time, the following must be done by the due date:

- Final `.pdf` file submitted on Gradescope

Be sure to **assign questions to pages** when you turn in your assignment on Gradescope. A .5 point deduction will be made any time the question is not assigned to a page within your PDF.

- Select the first page of your .PDF submission to be associated with the “Formatting” section.

Getting started

1. In Canvas, you'll find the template for this assignment (“Lab 04 Template”). Download the file and open it.
2. Save it in your class project folder as “`lab-04.qmd`”.
3. Open the document. Replace “Your Name Here” with your name in the YAML (the header of the document). Also, replace “Today's Date Here” with today's date. Filling in your name and date will count towards your “Formatting” score from here on out.
4. Render your document to make sure your YAML is rendering correctly.

💡 Tip for the Lab

Write all code and narrative in your Quarto file. Write all narrative in complete sentences. Throughout the assignment, you should periodically render your Quarto document to produce the updated html file.

Load the tidyverse

In the code chunk labeled `load-packages`, copy and paste the code below to load in the tidyverse package for today's lab:

```
library(tidyverse)
```

💡 Globally setting warnings and messages to false

Note that in your template, warnings and message have been **set to false globally** (i.e. across the entire document) by specifying these options in the YAML. Since we will be loading in packages and a dataset, this will be helpful for you formatting on this assignment.

It can be helpful to set global options this way in practice. However, take care when choosing this route, as sometimes warnings and messages can be helpful for troubleshooting.

Practice with ggplot and tidyverse

Today's lab involves more practice with ggplot and the tidyverse. You will also learn about loading data into R and setting a working directory to look for data.

Fast foods have been criticized as offering unhealthy foods which are calorically dense, high in sodium and added sugars, and low in vitamins and minerals. Today's dataset was compiled by Joey Stanley at BYU and consists of information regarding McDonald's menu items.

Best practices for data Visualization

When performing data visualization, make sure that all your plots have clear and informative titles and axis labels. When investigating more complex relationships with many variables, this simple tip will save you and your readers a lot of time and confusion.

Loading in data

Previously, you loaded data directly into R by using the `read.csv` command that pointed to a website. But what if you had a dataset on your computer?

You can still do so, but need to point R to reference the appropriate space on your computer. For instance, if I downloaded the McDonald's Menu Items dataset to my Downloads folder and named it `file.csv`, I might be able to load the dataset into R under the name `mcd` with the following command:

```
mcd <- read.csv("C:/Users/Kara/Downloads/file.csv")
```

Note that `file.csv`, the name of the file on your computer, could be different from the object name in R (this case, `mcd`).

However, file paths can be quite long. **There's a better way!** This is where our project folder comes in.

Navigate to your project folder

First, make sure you are in your project folder. You should see a blue cube at the top right of your RStudio window with the name of your project. If you see it, great. If not, you can open your project with File > Open Project.

At the beginning of class, you should have created a `data` folder that lives in your project folder. If you copy your data (perhaps it's called `file.csv`) into your `data` folder, you can read it in with the following code:

```
mcd <- read.csv("data/file.csv")
```

Because we are working out of our project folder, R knew to **start looking in the data folder within the project by default**. Using this strategy provides a shortcut instead of needing to provide the entire file path.

Any R output will go into your project folder (for instance, if you want to specifically save plots, figures, etc.) by default.

Note: there are some syntax issues with the way R parses in slashes, and this may change from Mac to PC (... anyone use Linux?). You can always check your current working directory with the following command in order to check the formatting of the file path (and also see what the current directory is!):

```
getwd()
```

```
[1] "C:/Users/kmcco/OneDrive - University of North Carolina at Chapel Hill/Documents/UNC_Teaching/fall2026/bios600-fa26/labs"
```

Downloading the data

First, download the Joey Stanley dataset. The data are available on his website here (<https://joeystanley.com/data/>), and clicking on “McDonald’s Menu Items. Save the data in your `data` folder within your project folder as `menu.csv`.

Exercise 1

a. In your code chunk labeled `load-data`, read in the data with the following code:

```
mcd <- read.csv("data/menu.csv")
```

Watch out for the naming of the file and where it’s saved on your computer! Print the first seven rows of the dataset with `slice()`.

⚠ Getting an error?

If you’re getting an error here, check the following:

- That you have your project open within RStudio (there is a blue cube at the top right of your RStudio window with the name of your project folder next to it. If you don’t see this, click File > Open Project... and navigate to your project folder.)
- That there is a folder called `data` within this project folder, and `menu.csv` is saved in that `data` folder.

If you’ve checked these things and are still getting an error, reach out to the teaching team for help!

Exercise 1

- Load in the McDonald’s dataset from Joey Stanley’s website (save the file to your computer).
- Print the column names in your dataset with `names()`.

On your own!

Exercise 2

Create a table which displays the number of menu items in each food category. (Hint: use the `count()` function.) Which category has the most number of items? Which category has the least?

Exercise 3

- a. Modify the code below to create a new variable called `cal_oz` which calculates the **calories per ounce** in each of the menu items. In this code, we are saving the dataset to a new object called `mcd_cal`.

```
mcd_cal <- mcd |>
  mutate(_____) # create the new variable here
```

- b. Create a new R Chunk. Now, within your newly created dataset `mcd_cal`, **select** the variables `Category`, `Item`, and `cal_oz`, then print the first five observations with `slice()`. Within the first five observations, which food item has the highest calories per ounce?

Exercise 4

- a. Create a table which displays the top five most caloric menu items (only based on total calories), with the highest-value item at the top of the table. In doing so, display only the name of the menu item and the calories. You may use the following code to get started:

```
mcd_cal |>
  select(____, ____)|>
  arrange(____)|> # arrange highest to lowest
  slice(____)
```

- b. Now, using similar code as above, create a second table which displays the top five most caloric menu items **per ounce** (based on the new variable you created in Exercise 2).
- c. Which item ranks the highest in each of the tables? Are you surprised?

Exercise 5

There are 9 calories per gram of fat. Create a new variable which corresponds to the proportion of calories that come from fat for each of the menu items. (Assign this dataset, with the newly created variable, to the object `mcd_cal`.) You can define the new variable as follows: `prop_cal_fat = Fat*9/Calories`. (Nothing needs to be printed out here. Just show your code.)

Exercise 6

Create a visualization that shows the distribution of calories in McDonald's menu items by category. Use the following code to get started, filling in the blanks. (You do not need to alter the `geom_boxplot` or the `theme()` parts of the code.)

```
ggplot(data = _____,
       mapping = aes(y = _____,
                     x = _____,
                     fill = _____)) +
  geom_boxplot() +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "_____")
```

What patterns do you see?

Exercise 7

Create a visualization that plots the calories per ounce of items (x-axis) by the proportion of calories that come from fat (y-axis). In this visualization, color by the food category. What pattern(s) do you see?

AI Attestation

If you used AI for this assignment, list your prompts below. Cite any other online resources you used. If you did not use AI for this assignment, you may simply state "I did not use AI for this assignment."

Submission

As you've seen previously, we can **Render** the template into an .html file that can be opened by any web browser. To export it as a .pdf, open the file in your web browser and then print to or save as a .pdf document. Your TAs will show you how if you need help! (There is a way to directly knit to a .pdf file, but it's quite a bit more involved.)

You will submit the PDF documents for labs and homework to Gradescope as part of your final submission.

To submit your assignment:

- Access Gradescope through the menu on the BIOS 600 Canvas site.
- Click on the assignment, and you'll be prompted to submit it.
- Mark the pages associated with each exercise. All of the pages of your lab should be associated with at least one question (i.e. should be "checked")

one question (i.e., should be checked).

- Select the first page of your .PDF submission to be associated with the “Formatting” section.

Grading

Component	Points
Ex 1	3
Ex 2	2
Ex 3	3
Ex 4	3
Ex 5	1
Ex 6	3
Ex 7	3
Formatting	3
Total	21

The “Formatting” grade is to assess the document format. This includes having a neatly organized document (no excessive output, warnings/messages when loading packages and/or data) with readable code and your name and the date updated in the YAML.